Python:

Python can be used to develop applications for frontend, backend, web scraping, scripting, cloud environments, AI, ML, data science, generative AI, Agentic AI, and AI agents.

Where we are not using:

- Python is not commonly used for developing mobile applications.
- But there is a Kivy framework

Basics:

- Web Application - The applications which will provide services over the web are called web applications.
    - Eg: gmail.com, facebook.com
- Every web application contains 2 main components
    - Front-End 2. Back-End
- Front-End - It represents what user is seeing on the website

Ex: HTML, JS, CSS, JQuery and BootStrap, React JS, Angular JS

- Back End:It is the technology used to decide what to show to the end user on the Front-End. i.e., Backend is responsible to generate required response to the end user, which is displayed by the Front-End.
- Back-End has 3 important components:

    1) The Languages like Java,Python etc

    2) The Framework like DJango,Pyramid,Flask etc

    3) The Database like SQLite,Oralce,MySQL etc

Web Framework

- A web framework is a collection of tools and libraries that help developers to build and maintain web applications. It's also known as a web application framework (WAF). -
- Benefits of using a web framework
    - Saves time: Developers don't need to write everything from scratch
    - Makes development more efficient: Developers can reuse code and follow a consistent development approach

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- Examples of web frameworks
    - Django, Flask, FastAPI,Bottle, Pyramid
    - Data Science Frameworks – Pandas, Numpy
    - ML/AI – TensorFlow, PyTorch
    - GUI – Tkinter

DJango

- Django is a high-level, free and open source Python web framework that helps developers build web applications quickly and securely.
- It provides:

  Built-in admin panel
  Authentication system (login/logout)
  ORM (database handling without SQL)
  Security features (SQL Injection, CSRF – Cross Site Req. Forgery)
  MVT architecture
  Fast development

- Initial Development:
    - 2003: Adrian Holovaty and Simon Willison – for newspaper content management - Lowrence Journal-World
    - The Django was named in the memory of Guitarist Django Reinhardt
    - Official website: djangoproject.com

Key Features

- Reusability – Promotes reusable code via it modular design – follows DRY (Don't Repeat Yourself) Principle.
- Rapid Development – The framework emphasizes fast development with a "batteries-included" i.e., a product comes with all the parts it needs to function
- Contains Object-Relational Mapping (ORM) - a programming technique that connects object-oriented programming language i.e., python objects to relational databases.
- Security – provides built-in protection mechanism – SAND Box, SQL Injection, CSRF…
- Scalability: Designed to handle high- traffic sites and large applications
- Versatility: Suitable for building everything from simple websites to complex enterprise solutions i.e., any kind of app we can develop.

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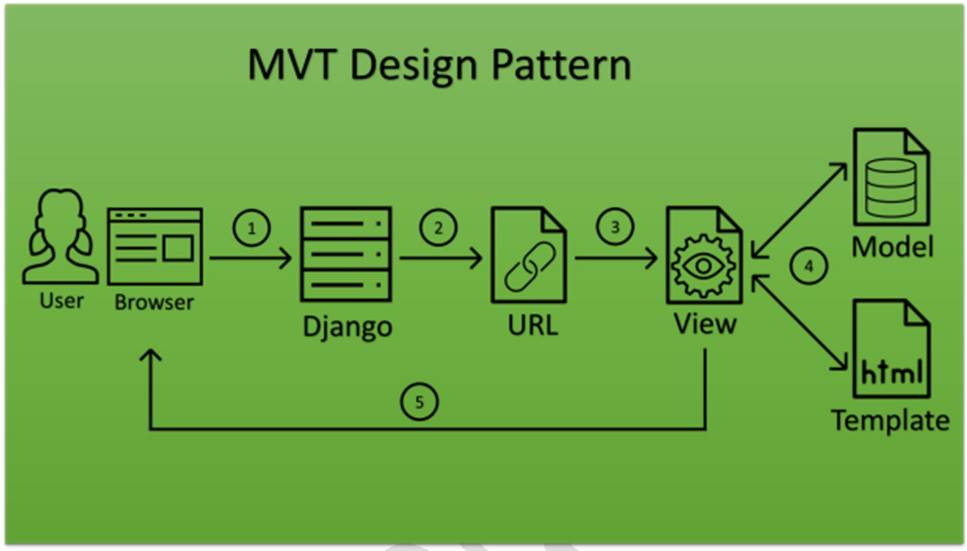
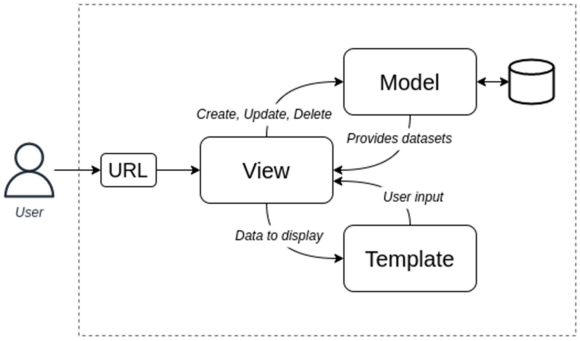
- The benefits of using Django for web development also include its Representational State Transfer (REST) framework - a popular toolkit for building web APIs. One of its key advantages is that it's extremely flexible: data is not tied to any methods or resources.

MVT Architecture:





Step-by-step:

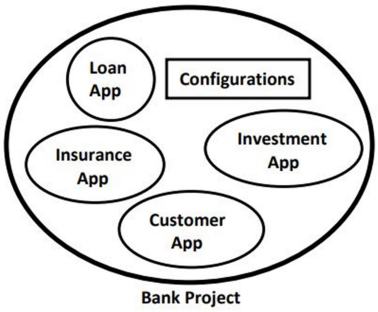
- Client sends request
- URL dispatcher calls the View
- View processes logic
- View sends data to Template
- Template renders HTML
- View returns HttpResponse to client

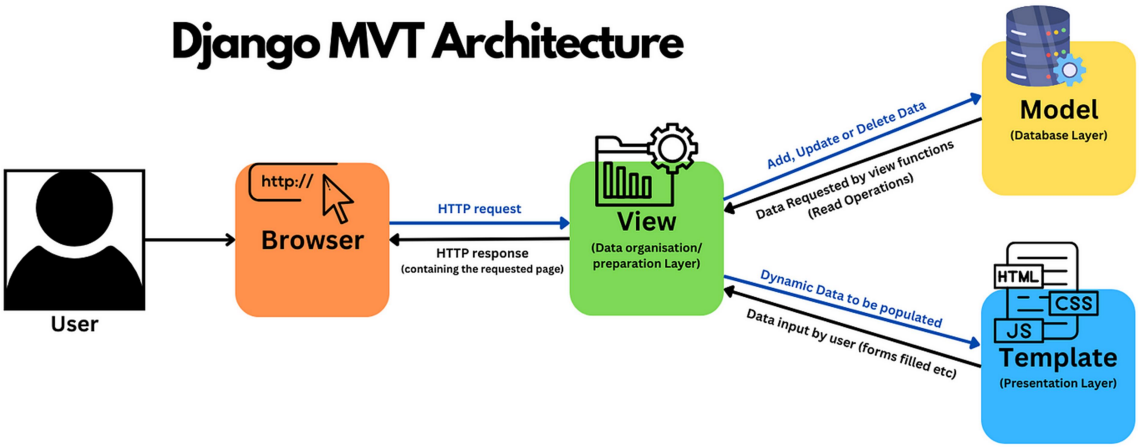
Client → URL → View → Model → View → Template → View → Client

- Models:
    - Defines database tables
    - Defines structure of data (fields and data types)
    - Represents data layer
    - Uses Django ORM (Object Relational Mapping)
    - Handles database interactions automatically
- Templates:
    - Controls how data is presented to the user.
    - Template is HTML page
    - Displays data dynamically
    - Handles presentation layer only
- Views:
    - Handles the logic of your application
    - Process the user request, interacts with models and returns response to the user by consulting with templates.

DJango Project vs DJango Application:

- A Django project is a collection of applications and configurations which forms a full web application.
    - Bank Project
- A Dango Application is responsible to perform a particular task.
    - loan app, registration app





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<u>Visual Studio Code Configuration for Python and DJango Development</u>

Step 1: Install Python (Version 3.O)

Step 2. Download and Install Visual Studio Code

Step 3:

- Open VS Code
- Click File → Open Folder
- Create a new folder like: F:\Courses\DJANGO\DJANGO 25-26\DJANGO_PROJECTS
- Open that folder Ex: DJANGO_PROJECTS

Step 4: Install Required VS Code Extensions

- Go to Extensions (Ctrl+Shift+X) and install:
  - Python (by Microsoft) ⭑ Mandatory
  - Pylance (auto installs usually)
  - Django (optional but helpful)

Step 5: Check Python is Connected to VS Code

- Press **Ctrl + Shift + P**
- Type:
  Python: Select Interpreter
- Select **Python 3.13**

Step 6: Create Virtual Environment (VERY IMPORTANT)

- Click Terminal → New Terminal
- Type:
  python -m venv name_of_the_environment
- Activate it:
  name_of_the_environment\Scripts\activate
- If successful, you'll see:
  (name_of_the_environment)

Note: If an error no module name venv came, check Python version

Step 7: Install DJango

- In the same terminal: type
  pip install Django or python –m pip install Django
- Check version:
  django-admin –version

Step 8: Create Django Project

- Inside same folder:
  django-admin startproject Name_of_the_Project

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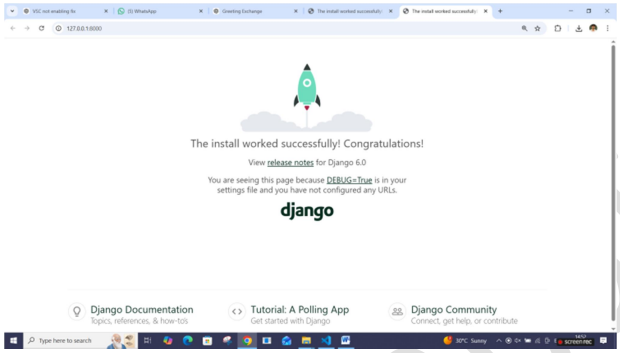
STEP 9: Run Django Server

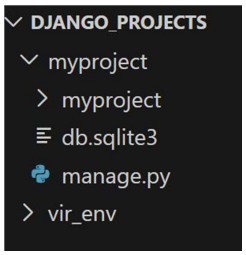
- cd Name_of_the_Project
- python manage.py runserver
- Open browser and type http://127.0.0.1:8000/. If you see Django rocket page, everything is perfectly configured.



Step 10: Optional (But Recommended)

- After creating virtual environment
- Again do:
  Ctrl + Shift + P
  Python: Select Interpreter
- Now select:
  ./venv/Scripts/python.exe
- This ensures VS Code uses your virtual environment.

Final Structure Should Look Like



Note: When we try to set the virtual environment, there is a chance of getting error related to execution policy restriction.

Error: Windows PowerShell does not allow running script files (.ps1)

Then type

Set-ExecutionPolicy -ExecutionPolicy Unrestricted -Scope CurrentUser

It allows script execution for your current user only.

Or

Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser

Then press **Y**

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After That

- Close PowerShell
  Open VS Code again
  Open Terminal
- Try: name_of_the_environment \Scripts\activate
- It should work now and show:

  (name_of_the_environment)

To stop (deactivate) virtual environment, just type: deactivate

PIP Installation:

- Open the Command Prompt:
- Press Win + R, type cmd, and press Enter.
- Check Python Installation:
- Run:

  python --version
- Ensure Python is installed and the path is configured.
- Install pip with ensurepip:

  python -m ensurepip –upgrade

When the project is created in Workspace folder the following structure will be created:

__init__.py: It is a blank python script. Because of this special file name, Django treated this folder as python package.

settings.py: Database settings, Security settings, Application Settings

urls.py: Maps URL to view

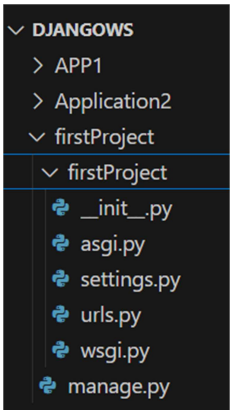
wsgi.py: wsgi - > Web Server Gateway Interface.

Helps communicate with web servers like

Gunicorn

Apache HTTP Server

Used in production environment.



manage.py: Run development servers, Create apps, Apply migrations Open Django shell

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asgi.py : Used for asynchronous (The system can handle multiple tasks at the same time without waiting for one task to finish before starting another) support. It supports Real-time applications.

When the Application is created in Project folder the following structure will be created:

- **models.py** - Defines database tables.
- **views.py** - Handles business logic.
- **admin.py** - Registers models for Django Admin panel.
- **tests.py** - Used for testing.
- **migrations/** - Stores database change history.

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